

Replication Report:
*Ferroelectric control of magnetic skyrmions in two-dimensional
van der Waals heterostructures*
Huang, Shao & Tsymbal — arXiv:2202.11348

REPLICATE-PROJECT / TEXTURES-100 (textures-polar-huang2022)

Independent reimplementations — no author code

Abstract

We independently reimplement the core physics of Huang *et al.* (arXiv:2202.11348): ferroelectric (FE) control of magnetic skyrmions in $\text{Fe}_3\text{GeTe}_2/\text{In}_2\text{Se}_3$ van der Waals heterostructures. The paper predicts, from DFT + atomistic spin-dynamics, that reversing the In_2Se_3 FE polarization switches the interfacial Dzyaloshinskii–Moriya interaction (DMI) between $D_\uparrow = 0.28$ and $D_\downarrow = 0.06$ mJ/m², thereby *creating* vs. *annihilating* a ~ 12 nm Néel skyrmion, and that in the trilayer the DMI sign flips ($D_{\uparrow\uparrow} \approx 0.22$, $D_{\downarrow\downarrow} \approx -0.24$ mJ/m²) reversing skyrmion chirality. We rebuild a discrete square-lattice micromagnetic model (exchange + interfacial Néel DMI + uniaxial anisotropy) and relax a circular seed by overdamped LLG. The analytic critical DMI, $D_c = (4/\pi)\sqrt{AK} = 0.255$ mJ/m², brackets the paper’s cases exactly: $D_\uparrow = 0.28 > D_c > D_\downarrow = 0.06$, reproducing the create/annihilate switch with zero fitting. Topological charge $|Q| \approx 1$ confirms genuine skyrmions; larger $|D|$ yields larger skyrmions (trend reproduced).

VERDICT: PARTIAL (leaning REPLICATED on mechanism). Coverage $\approx 6\text{--}7/10$, Agreement $\approx 7/10$.
Core mechanism (DMI-controlled skyrmion creation/annihilation via FE switching, and chirality tied to $\text{sign}(D)$) is **fully reproduced**. Absolute skyrmion diameter is **off by $\sim 2\text{--}3\times$** because the exchange stiffness A appears only in the paper’s Fig. 4 (not in extractable text); diameter $\propto \sqrt{A}$.

1 The paper’s claims

1. **FE-switchable DMI (bilayer).** DFT gives $D_\uparrow = 0.28$ mJ/m² (polarization up) and $D_\downarrow = 0.06$ mJ/m² (down).
2. **Create / annihilate switch.** With adopted anisotropy $K \approx 0.04$ MJ/m³, a ~ 12 nm Néel skyrmion emerges for $D_\uparrow$ and *disappears* for $D_\downarrow$ — controlled entirely by FE polarization.
3. **Sign flip & chirality reversal (trilayer).** $D_{\uparrow\uparrow} \approx 0.22$ mJ/m², $D_{\downarrow\downarrow} \approx -0.24$ mJ/m² with $D_{\uparrow\uparrow} \approx -D_{\downarrow\downarrow}$ and $D_{\uparrow\downarrow} \approx D_\uparrow - D_\downarrow$; a ~ 6 nm skyrmion whose chirality reverses with polarization.
4. **Anisotropy caveat.** The DFT MAE (8.57 MJ/m³) is too large (would give $R \sim 0.03$ nm); the authors adopt a moderate $K \approx 0.04$ MJ/m³ to qualitatively demonstrate the FE effect.

2 Governing equations reimplemented

Spin Hamiltonian (paper Eq. 5):

$$\mathcal{H} = -\sum_{\langle ij \rangle} J \hat{\mathbf{m}}_i \cdot \hat{\mathbf{m}}_j + \sum_{\langle ij \rangle} \mathbf{D}_{ij} \cdot (\hat{\mathbf{m}}_i \times \hat{\mathbf{m}}_j) - \sum_i K (\hat{\mathbf{e}} \cdot \hat{\mathbf{m}}_i)^2, \quad (1)$$

with interfacial (Néel) DMI $\mathbf{D}_{ij} = D (\hat{\mathbf{z}} \times \hat{\mathbf{u}}_{ij})$. Relaxation by overdamped LLG (projected gradient), $\dot{\hat{\mathbf{m}}} = \hat{\mathbf{m}} \times (\hat{\mathbf{m}} \times \mathbf{H}_{\text{eff}})$, $\mathbf{H}_{\text{eff}} = -\partial \mathcal{H} / \partial \hat{\mathbf{m}}$, normalizing each step.

Expected skyrmion radius (paper Eq. 6):

$$R = \Delta \frac{\pi D}{16\sqrt{AK} - \pi D}, \quad \Delta = \sqrt{A/K}. \quad (2)$$

Verdict anchor — the critical DMI (spiral/skyrmion instability threshold), equivalent to the Eq. 6 denominator vanishing up to an $O(1)$ prefactor:

$$\boxed{D_c = \frac{4}{\pi} \sqrt{AK}}. \quad (3)$$

A Néel skyrmion is stabilized when $|D| \gtrsim D_c$.

3 Methods

Discrete square lattice, 60×60 cells at 1 nm spacing (matching the paper's 60×60 nm supercell). Energy terms: exchange $A/dx^2 \sum (\hat{\mathbf{m}}_i - \hat{\mathbf{m}}_j)^2$, uniaxial anisotropy $-K \sum m_z^2$, and interfacial Néel DMI as bond terms $\mathbf{D}_{ij} \cdot (\hat{\mathbf{m}}_i \times \hat{\mathbf{m}}_j)$ with $d = D dx$. A circular reversed-core seed (radius $r_0 = 8$ nm, smooth 180° wall) is relaxed for 5000 overdamped-LLG steps. **Seed handedness tracks sign(D)**: $\phi = \text{atan2}(Y, X) + (0 \text{ if } D < 0 \text{ else } \pi)$, required so the initial chirality does not oppose the DMI and spuriously collapse the seed.

Fixed parameters. $A = 1.0$ pJ/m (typical Fe_3GeTe_2 ; the paper's value is only in Fig. 4, not extractable from text), $K = 0.04$ MJ/m³ (paper's adopted modelling value), $dx = 1$ nm.

Diagnostics. Berg-Lüscher lattice topological charge $Q = \frac{1}{4\pi} \sum \hat{\mathbf{m}} \cdot (\hat{\mathbf{m}}_{x+} \times \hat{\mathbf{m}}_{y+})$; skyrmion diameter from reversed-core area, $2\sqrt{\text{area}/\pi}$ where $m_z < 0$. Interpreter: `/home/stevens/comfyui-env/bin/python` (numpy 2.3.5).

4 Results

Live re-run reproduces the saved evidence JSON to the quoted digits. Analytic $D_c = 0.2546$ mJ/m².

5 Comparison & verdict

The single most important result is that the closed-form $D_c = (4/\pi)\sqrt{AK} = 0.255$ mJ/m² lies *between* the paper's two bilayer DMI values, $D_\downarrow = 0.06 < D_c < D_\uparrow = 0.28$ mJ/m². This reproduces the paper's central create/annihilate switch **analytically and with zero fitting**: the FE polarization state, acting through the DMI it induces, is what turns the skyrmion on and off. The

Case	D (mJ/m ²)	vs. D_c	Skyrmion? (this work)	Q	Diameter (nm)
$D_\uparrow$	+0.28	$> D_c$	yes	-0.87	39.5
$D_\downarrow$	+0.06	$< D_c$	no	0.0	0.0
$D_{\uparrow\uparrow}$	+0.22	$< D_c^\dagger$	yes	-0.90	19.9
$D_{\downarrow\downarrow}$	-0.24	$ D < D_c^\dagger$	yes	-0.99	28.2

Table 1: This-work relaxation results. [†]The trilayer cases sit just below the analytic D_c but still relax to a metastable skyrmion from the seed (Eq. 6 predicts a finite though small R as $|D| \rightarrow D_c$); the seeded overdamped relaxation captures the metastable branch.

Quantity	This work	Paper
Create/annihilate switch ($D_\uparrow$ vs $D_\downarrow$)	reproduced	core claim
Skyrmion present at $D_\uparrow = 0.28$	yes, $Q = -0.87$	yes (~ 12 nm)
Skyrmion absent at $D_\downarrow = 0.06$	yes (none)	none
Chirality set by $\text{sign}(D)$	reproduced (seed)	yes (trilayer)
Size trend (larger $ D \rightarrow$ larger)	reproduced	12 vs 6 nm
Absolute diameter (bilayer up)	39.5 nm	~ 12 nm
Critical DMI D_c	0.255 mJ/m ²	implied by Eq. 6

Table 2: This-work vs. paper. Qualitative/mechanistic claims match; absolute diameter is $\sim 2\text{--}3\times$ too large (see gaps).

seeded relaxation confirms this dynamically ($Q = -0.87$ for $D_\uparrow$; seed collapses for $D_\downarrow$), and the Néel handedness follows $\text{sign}(D)$ as the paper’s trilayer chirality-reversal claim requires. The size trend (larger $|D| \rightarrow$ larger skyrmion) is also reproduced. We therefore judge the **core mechanism REPLICATED**, with an overall **PARTIAL** verdict driven solely by the absolute-size mismatch below.

6 Honest gaps & scope

1. **Exchange stiffness A unknown (absolute size).** A is reported only in the paper’s Fig. 4 (an image), not in the OCR/pdf-to-text-extractable text. We used the typical Fe₃GeTe₂ value $A = 1$ pJ/m. Since skyrmion size $\propto \sqrt{A}$ (and $D_c \propto \sqrt{A}$), our ~ 40 nm diameter is $\sim 2\text{--}3\times$ the paper’s ~ 12 nm; the *threshold and trend are correct*, the absolute scale is uncertain by $\sqrt{A_{\text{assumed}}/A_{\text{true}}}$.
2. **Square lattice, not the true Fe honeycomb.** The paper keeps the Fe honeycomb; we approximate with a square lattice, which slightly changes the geometric DMI prefactor but not the mechanism.
3. **FE polarization represented via DMI proxy.** The paper’s atomistic model has no explicit polarization field — FE state enters only through the DMI magnitude/sign it induces (via DFT). We correctly sweep D as that proxy rather than inventing a polarization term.
4. **No demagnetizing/dipolar field** (thin-film 2D limit, as the paper’s atomistic model also omits it).

5. **Diameter measured via reversed-core area** ($m_z < 0$), a proxy for the paper’s domain-wall definition.

7 Reproducibility

```
# Interpreter with numpy 2.3.5 / scipy:  
/home/stevens/comfyui-env/bin/python work/skymion.py  
# -> prints Dc and per-case {diameter, Q, skymion}; writes work/_sim_out.json  
# Evidence copies: report/evidence/{skymion.py, huang2022_result.json, _sim_out.json}
```

Note: no LaTeX engine on the packaging host; this `.tex` is delivered as source and compiles off-host (`pdflatex REPORT.tex`). Extraction artifacts (`extraction/marker.md`, `extraction/nougat.mmd`) are pdftotext interims (`marker/nougat` binaries absent) — an environment gap, not a physics gap.